

Questions

Q1. Convert to English (10 points)

Explain the following relational algebra queries in plain English. Be precise in your wording and include terms like "all," "every," or "at least one" where appropriate. If a query cannot be expressed, clearly state the reason.

1. $\pi_{\text{name}} (\sigma_{\text{city} = \text{'Toronto'}} (\text{Buildings} \bowtie \text{Rooms} \bowtie \text{Sensors} \bowtie \text{Readings})) \cap \pi_{\text{name}} (\sigma_{\text{height} > 150} (\text{Buildings} \bowtie \text{Supplies} \bowtie \text{Suppliers}))$
2. $\pi_{\text{room_id}} (\sigma_{\text{capacity} > 100} (\text{Rooms} \bowtie \text{Buildings})) - \pi_{\text{room_id}} (\sigma_{\text{status} = \text{'inactive'}} (\text{Sensors} \bowtie \text{Readings}))$
3. $\pi_{\text{person_id}, \text{maintenance_id}, \text{task_id}, \text{task_type}} (\text{Tasks} \bowtie \text{PersonnelAssignment} \bowtie \text{Personnel}) \div \{ \{ \text{task_type:string} \}$
 "Inspection"
 "Maintenance"
 $\}$
4. $\pi_{\text{supplier_id}} ((\sigma_{\text{cost} < 100} (\text{Supplies} \bowtie \text{Buildings} \bowtie \text{Tasks})) \cap (\sigma_{\text{city} = \text{'Vancouver'}} (\text{Suppliers})))$
5. $\pi_{\text{name}} (\text{Buildings} \bowtie (\pi_{\text{building_id}} (\sigma_{\text{task_type} = \text{'Cleaning'} \wedge \text{priority} = \text{'High'}} (\text{Tasks} \bowtie \text{Supplies})) \cap \pi_{\text{building_id}} (\sigma_{\text{task_type} = \text{'Maintenance'} \wedge \text{priority} = \text{'Low'}} (\text{Tasks} \bowtie \text{PersonnelAssignment} \bowtie \text{Personnel}))))$
6. $\pi_{\text{room_id}} ((\sigma_{\text{floor} = 1} (\text{Rooms})) \bowtie (\pi_{\text{building_id}} (\sigma_{\text{height} > 100} (\text{Buildings}))))$
7. $\pi_{\text{sensor_id}} (\sigma_{\text{status} = \text{'active'}} (\text{Sensors} \bowtie \text{Readings})) - \pi_{\text{sensor_id}} (\sigma_{\text{type} = \text{'temperature'}} (\text{Sensors} \bowtie \text{Rooms} \bowtie \text{Buildings}))$
8. $\pi_{\text{Buildings_renamed.building_id}, \text{Buildings_renamed.name}} (\sigma_{\text{Rooms.capacity} > 100 \wedge \text{Sensors.status} = \text{'active'}} (\rho_{\text{Buildings_renamed}} (\text{Buildings}) \bowtie_{\text{Rooms.building_id} = \text{Buildings_renamed.building_id}} \text{Rooms} \bowtie_{\text{Sensors.room_id} = \text{Rooms.room_id}} \text{Sensors}))$
9. $\pi_{\text{Buildings_renamed.building_id}, \text{Buildings_renamed.name}} (\sigma_{\text{Rooms.capacity} > 100 \wedge \text{Sensors.status} = \text{'active'}} (\rho_{\text{Buildings_renamed}} (\text{Buildings}) \bowtie_{\text{Rooms.building_id} = \text{Buildings_renamed.building_id}} \text{Rooms} \bowtie_{\text{Sensors.room_id} = \text{Rooms.room_id}} \text{Sensors}))$
10. $\pi_{\text{name}} (\text{Buildings} \bowtie (\pi_{\text{building_id}} (\text{Buildings} \bowtie \text{Rooms} \bowtie \text{Sensors}) \div \pi_{\text{building_id}} (\sigma_{\text{height} > 300} (\text{Buildings} \bowtie \text{Supplies} \bowtie \text{Tasks}))))$

Q2. Relational Algebra Queries (36 points)

Provide **relational algebra queries** for the following tasks:

If a task cannot be expressed in relational algebra, write "Cannot be done in RA" and justify why.

1. Retrieve the names of buildings that have at least one active sensor.
2. Retrieve the IDs of buildings with rooms on every floor between 1 and 10 inclusive.
3. Retrieve the IDs of buildings with tasks of all priority levels (Assume there are three priority levels: High, Medium, Low).

4. Assume there are three buildings with IDs 1, 2, and 3 among other buildings. Find the names of buildings out of the three where every room has at least one sensor installed.
5. Retrieve the IDs of suppliers providing resources to every building in Toronto.
6. Retrieve the names of suppliers who supply resources to all buildings that have high-priority tasks assigned.
7. Retrieve all rooms that have at least two sensors of different types installed.
8. Find all buildings that have both inactive sensors and a scheduled maintenance event in the same month.
9. Retrieve the IDs of rooms in Toronto where the total capacity of two rooms exceeds 500. Each pair should have unique rooms.
10. Retrieve the ID of the tallest building such that sensors installed in rooms on every floor. Assume there is only one tallest building.
11. Find the names of personnel responsible for maintenance tasks in buildings where resources were supplied by suppliers located in the same city, and the building has at least one high-priority task assigned.
12. Retrieve the IDs of buildings where at least one sensor has recorded a reading above 75 after 23rd October 2024, the buildings have had maintenance tasks in the same period, and they received supplies from suppliers located in a different city.

Q3. SQL Queries (36 points)

For each part in Q2, write the corresponding **SQL query**. Ensure that:

- The schema matches the one provided above.
- Your queries execute correctly in SQLite.

Save all your RA queries in a file named **A1.txt** and all SQL queries in a file named **A1.sql**.

Q4. Primary Key and Foreign Key Relationships (5 points)

Identify the **primary key** and **foreign key** relationships for each schema defined above.

Q5. Revise the Schema (2 points)

In your opinion, this database schema is bad. Show off your skills to the manager and rewrite the schema. There are no right or wrong answers. Marks are based on the quality of answers, and any theories, guidelines or practices you've incorporated. Feel free to take some time to research the topic but remember to cite your sources.

Provide:

1. The **revised schema**.
2. **Reasons** for your decisions.

Q6. Bonus (1 point)

Create a **funny meme or short story** about your interaction with the smart city manager. If it makes the TA grin, you'll get the bonus mark!

Styling marks (5 points)

Styling of your submission will be evaluated and carries 4 points.

Please follow submission instructions mentioned at the beginning.

Submit your assignment (A1.txt, A1.sql, A1.pdf) on MarkUs.